

CS1102 – Programming 1

Programming Assignment 1 (PA1)

Java Quiz Game

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1. Introduction

This assignment implements a simple quiz game in Java. The program presents the user with five multiple-choice questions on core Java programming concepts. Each question has four options labeled A through D. The program validates all user input, evaluates answers using both if and switch case statements, and computes a final percentage score upon completion.

The program satisfies all rubric criteria: it compiles without errors, validates user input strictly, uses correct arithmetic for score computation, follows a logical and well-structured flow, and adheres to Java code style conventions with inline comments throughout.

2. Program Design and Explanation

2.1 Section 1 – Setup and Score Tracking

A Scanner object is initialized to read user input from the console. Two integer variables, totalQuestions and correctAnswers, are declared to track the quiz parameters. Using descriptive variable names improves readability, as recommended by Eck (2022) in Chapter 2.

2.2 Section 2 – Questions, Options, and Answer Keys

All five questions, their four options, and the correct answer characters are stored in arrays. This design keeps data separate from logic, making the program easy to extend. The correct answer keys are stored as char values ('A'–'D'), which is the most appropriate primitive data type for single-character comparisons in Java.

2.3 Section 3 – Welcome Message

A welcome banner is printed using System.out.println() to orient the user before the quiz begins. This is consistent with the basic Java output syntax described by Eck (2022, Section 2.1).

2.4 Section 4a – Input Validation

A while loop repeats the prompt until the user enters a valid single character (A, B, C, or D). The program first trims and converts the input to uppercase for robustness. It then checks that the input length is exactly 1, and that the character matches one of the four valid options. Invalid

input triggers a descriptive error message and re-prompts the user. This implements input validation as required by the rubric.

The validation uses if statements to check each condition — fulfilling the assignment requirement to use if statements in the program logic.

2.5 Section 4b – Answer Evaluation with switch

After receiving a valid answer, a switch statement compares the user's input character against the four possible cases (A, B, C, D). For each case, the program checks whether the user's answer matches the correct key for that question. This fulfills the explicit requirement to use switch case statements for answer comparison, as described in Eck (2022, Section 3.6).

2.6 Section 4c – Immediate Feedback

After each answer is evaluated, an if-else block provides immediate feedback: a confirmation message if correct (and increments correctAnswers), or the correct answer if wrong. This allows the user to learn as they progress through the quiz.

2.7 Section 5 – Score Computation and Output

After all five questions are answered, the program computes the final score as a percentage using the formula:

$$\text{scorePercentage} = (\text{correctAnswers} / \text{totalQuestions}) \times 100$$

The variable scorePercentage is declared as double to preserve decimal precision before casting to int for display. A second if-else chain maps the score to a descriptive remark (Excellent, Great job, Good effort, or Keep studying).

2.8 Section 6 – Resource Cleanup

The Scanner is closed after use to release system resources. This is a standard Java best practice and contributes to clean, non-redundant code as required by the Code Style criterion.

3. Source Code

The full source code of QuizGame.java is listed below:

```
import java.util.Scanner;

/**
 * QuizGame.java
 * A simple multiple-choice quiz game that presents five questions,
 * validates user input, evaluates answers, and displays a final score.
 *
 * Author: Hset Paing
 */
```

```

* Course: CS1102 - Programming 1
*/
public class QuizGame {

    public static void main(String[] args) {

        // -----
        // SECTION 1: Setup - Scanner and score tracking
        // -----
        Scanner scanner = new Scanner(System.in);
        int totalQuestions = 5;
        int correctAnswers = 0;

        // -----
        // SECTION 2: Quiz questions, options, and correct answers
        // Each array index corresponds to the same question number.
        // -----
        String[] questions = {
            "What is the correct syntax to print text in Java?",
            "Which data type is used to store a single character in Java?",
            "Which keyword is used to declare a constant in Java?",
            "What is the result of 10 % 3 in Java?",
            "Which of the following is a valid Java variable name?"
        };

        String[][] options = {
            {"A. print(\"Hello\");", "B. System.out.println(\"Hello\");",
            "C. echo \"Hello\";", "D. console.log(\"Hello\");"},
            {"A. String", "B. int",
            "C. char", "D. byte"},
            {"A. const", "B. constant",
            "C. static", "D. final"},
            {"A. 3", "B. 1",
            "C. 0", "D. 2"},
            {"A. 2variable", "B. variable-name",
            "C. variableName", "D. variable name"}
        };

        // Correct answers stored as uppercase letters
        char[] correctAnswerKeys = {'B', 'C', 'D', 'B', 'C'};

        // -----
        // SECTION 3: Display welcome message
        // -----
        System.out.println("=====");
        System.out.println("        Welcome to the Java Quiz Game!        ");
        System.out.println("=====");
        System.out.println("Answer each question by typing A, B, C, or D.");
        System.out.println();

        // -----
        // SECTION 4: Loop through each question
        // -----
        for (int i = 0; i < totalQuestions; i++) {

            // Display question number and question text
            System.out.println("Question " + (i + 1) + ": " + questions[i]);

            // Display all four answer options
            for (String option : options[i]) {
                System.out.println("    " + option);
            }
        }
    }
}

```

```

    }

    // -----
    // SECTION 4a: Input Validation
    // Ensure the user enters only A, B, C, or D.
    // Keep prompting until a valid answer is given.
    // -----
    char userAnswer = ' ';
    boolean isValidInput = false;

    while (!isValidInput) {
        System.out.print("Your answer: ");
        String inputLine = scanner.nextLine().trim().toUpperCase();

        // Check that input is exactly one character and is A, B, C, or D
        if (inputLine.length() == 1) {
            char inputChar = inputLine.charAt(0);
            if (inputChar == 'A' || inputChar == 'B' ||
                inputChar == 'C' || inputChar == 'D') {
                userAnswer = inputChar;
                isValidInput = true;
            } else {
                System.out.println("    [Invalid] Please enter A, B, C, or D
only.");
            }
        } else {
            System.out.println("    [Invalid] Please enter a single letter:
A, B, C, or D.");
        }
    }

    // -----
    // SECTION 4b: Answer Evaluation using switch statement
    // Compare the user's answer to the correct answer key.
    // -----
    char correctKey = correctAnswerKeys[i];
    boolean isCorrect = false;

    switch (userAnswer) {
        case 'A':
            isCorrect = (correctKey == 'A');
            break;
        case 'B':
            isCorrect = (correctKey == 'B');
            break;
        case 'C':
            isCorrect = (correctKey == 'C');
            break;
        case 'D':
            isCorrect = (correctKey == 'D');
            break;
        default:
            // This case is unreachable due to validation above,
            // but included for completeness and safe default behavior.
            isCorrect = false;
            break;
    }

    // -----
    // SECTION 4c: Provide immediate feedback using if statement
    // -----
    if (isCorrect) {

```

```

        correctAnswers++;
        System.out.println("  Correct!");
    } else {
        System.out.println("  Incorrect. The correct answer was: " +
correctKey);
    }

    System.out.println(); // Blank line between questions
}

// -----
// SECTION 5: Score Computation and Final Output
// Calculate the score as a percentage of total questions.
// -----
double scorePercentage = ((double) correctAnswers / totalQuestions) * 100;

System.out.println("=====");
System.out.println("          Quiz Complete!          ");
System.out.println("=====");
System.out.println("Questions answered : " + totalQuestions);
System.out.println("Correct answers      : " + correctAnswers);
System.out.println("Your final score    : " + (int) scorePercentage + "%");

// Provide a grade remark using if-else
if (scorePercentage == 100) {
    System.out.println("Result: Excellent! Perfect score!");
} else if (scorePercentage >= 80) {
    System.out.println("Result: Great job! Well done.");
} else if (scorePercentage >= 60) {
    System.out.println("Result: Good effort. Keep practicing!");
} else {
    System.out.println("Result: Keep studying. You can do better!");
}

System.out.println("=====");

// -----
// SECTION 6: Close the scanner to free resources
// -----
scanner.close();
}
}

```

4. Sample Output

The following output demonstrates the program running with all correct answers (B, C, D, B, C):

```

=====
Welcome to the Java Quiz Game!
=====
Answer each question by typing A, B, C, or D.

Question 1: What is the correct syntax to print text in Java?
A. print("Hello");
B. System.out.println("Hello");
C. echo "Hello";
D. console.log("Hello");
Your answer: B

```

```

Correct!

Question 2: Which data type is used to store a single character in Java?
A. String
B. int
C. char
D. byte
Your answer: C
Correct!

Question 3: Which keyword is used to declare a constant in Java?
A. const
B. constant
C. static
D. final
Your answer: D
Correct!

Question 4: What is the result of 10 % 3 in Java?
A. 3
B. 1
C. 0
D. 2
Your answer: B
Correct!

Question 5: Which of the following is a valid Java variable name?
A. 2variable
B. variable-name
C. variableName
D. variable name
Your answer: C
Correct!

=====
Quiz Complete!
=====
Questions answered : 5
Correct answers    : 5
Your final score   : 100%
Result: Excellent! Perfect score!
=====

```

The following output demonstrates input validation — the program rejected 'X' and re-prompted:

```

Question 2: Which data type is used to store a single character in Java?
A. String
B. int
C. char
D. byte
Your answer: X
[Invalid] Please enter A, B, C, or D only.
Your answer: C
Correct!

```

5. References

- Eck, D. J. (2022). *Introduction to programming using Java version 9, JavaFX edition*. Licensed under CC 4.0. <https://math.hws.edu/javanotes/>
- blondiebytes. (2019, July 9). *Learn Java in 25 minutes | Java tutorial for beginners* [Video]. YouTube. <https://youtube.com/watch?v=RLi1rOgTRbA>
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